

Mathematics for Information Technology

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Room.518 or Room.506 (MIV Lab)

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INTEGRATION (I)

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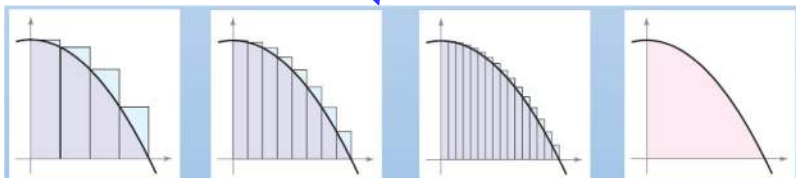
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Integration

• OBJECTIVES

How to

- I** – evaluate **indefinite integrals** using **basic integration rules**.
- II** – evaluate a **sum** and approximate the **area** of a plane region.



The area of a parabolic region can be approximated as the sum of the areas of rectangles. As you increase the number of rectangles, the approximation tends to become more and more accurate. you will learn how the limit process can be used to find areas of a wide variety of regions.

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Integration

• OBJECTIVES

How to

- evaluate a **definite integral** using a **limit**.
- III** – evaluate a **definite integral** using the **Fundamental Theorem of Calculus**.
- IV** – evaluate different **types** of definite and indefinite integrals using a variety of methods.

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Antiderivatives and Indefinite Integration

I

Objectives

- Write the general solution of a differential equation.
- Use indefinite integral notation for antiderivatives.
- Use basic integration rules to find antiderivatives.
- Find a particular solution of a differential equation.

EXAMPLE 1 Solving a Differential Equation

Find the general solution of the differential equation $y' = 2$.

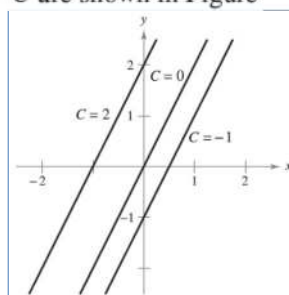
Solution To begin, you need to find a function whose derivative is 2. One such function is

$$y = 2x. \quad 2x \text{ is an antiderivative of } 2.$$

Now, you can use Theorem 4.1 to conclude that the general solution of the differential equation is

$$y = 2x + C. \quad \text{General solution}$$

The graphs of several functions of the form $y = 2x + C$ are shown in Figure



Antiderivatives

DEFINITION OF ANTIDERIVATIVE

A function F is an **antiderivative** of f on an interval I if $F'(x) = f(x)$ for all x in I .

Note that F is called an antiderivative of f , rather than the antiderivative of f . To see why, observe that

$$F_1(x) = x^3, \quad F_2(x) = x^3 - 5, \quad \text{and} \quad F_3(x) = x^3 + 97$$

are all antiderivatives of $f(x) = 3x^2$. In fact, for any constant C , the function given by $F(x) = x^3 + C$ is an antiderivative of f .

the **general solution** of the differential equation

THEOREM 4.1 REPRESENTATION OF ANTIDERIVATIVES

If F is an antiderivative of f on an interval I , then G is an antiderivative of f on the interval I if and only if G is of the form $G(x) = F(x) + C$, for all x in I where C is a constant.

general antiderivative of f

constant of integration

Notation for Antiderivatives

When solving a differential equation of the form

$$\frac{dy}{dx} = f(x)$$

it is convenient to write it in the equivalent differential form

$$dy = f(x) dx.$$

The operation of finding all solutions of this equation is called **antidifferentiation** (or **indefinite integration**) and is denoted by an integral sign $\int$. The general solution is denoted by

$$y = \int f(x) dx = F(x) + C.$$

Diagram labels: Variable of integration (x), Constant of integration (C), Integrand (f(x)), An antiderivative of f(x) (F(x)).

The expression $\int f(x) dx$ is read as the antiderivative of f with respect to x . So, the differential dx serves to identify x as the variable of integration. The term **indefinite integral** is a synonym for antiderivative.

Basic Integration Rules

The inverse nature of integration and differentiation can be verified by substituting $F'(x)$ for $f(x)$ in the indefinite integration definition to obtain

$$\int F'(x) dx = F(x) + C.$$

Integration is the “inverse” of differentiation.

Moreover, if $\int f(x) dx = F(x) + C$, then

$$\frac{d}{dx} \left[\int f(x) dx \right] = f(x).$$

Differentiation is the “inverse” of integration.

These two equations allow you to obtain integration formulas directly from differentiation formulas, as shown in the following summary.

Basic Integration Rules

Differentiation Formula

$$\frac{d}{dx}[C] = 0$$

$$\frac{d}{dx}[kx] = k$$

$$\frac{d}{dx}[kf(x)] = kf'(x)$$

$$\frac{d}{dx}[f(x) \pm g(x)] = f'(x) \pm g'(x)$$

$$\frac{d}{dx}[x^n] = nx^{n-1}$$

Integration Formula

$$\int 0 dx = C$$

$$\int k dx = kx + C$$

$$\int kf(x) dx = k \int f(x) dx$$

$$\int [f(x) \pm g(x)] dx = \int f(x) dx \pm \int g(x) dx$$

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C, \quad n \neq -1 \quad \text{Power Rule}$$

NOTE Note that the Power Rule for Integration has the restriction that $n \neq -1$.

THEOREM 5.5 LOG RULE FOR INTEGRATION

Let u be a differentiable function of x .

$$1. \int \frac{1}{x} dx = \ln|x| + C \quad 2. \int \frac{1}{u} du = \ln|u| + C$$

Basic Integration Rules

$$\frac{d}{dx}[\sin x] = \cos x$$

$$\int \cos x dx = \sin x + C$$

$$\frac{d}{dx}[\cos x] = -\sin x$$

$$\int \sin x dx = -\cos x + C$$

$$\frac{d}{dx}[\tan x] = \sec^2 x$$

$$\int \sec^2 x dx = \tan x + C$$

$$\frac{d}{dx}[\sec x] = \sec x \tan x$$

$$\int \sec x \tan x dx = \sec x + C$$

$$\frac{d}{dx}[\cot x] = -\csc^2 x$$

$$\int \csc^2 x dx = -\cot x + C$$

$$\frac{d}{dx}[\csc x] = -\csc x \cot x$$

$$\int \csc x \cot x dx = -\csc x + C$$

$$\int 3x dx = 3 \int x dx = 3 \left(\frac{x^2}{2} + C \right) = \frac{3}{2}x^2 + 3C.$$

EXAMPLE 2 Applying the Basic Integration Rules

Describe the antiderivatives of $3x$.

Solution $\int 3x dx = 3 \int x dx$ Constant Multiple Rule

$$= 3 \int x^1 dx$$

Rewrite x as x^1 .

$$= 3 \left(\frac{x^2}{2} \right) + C$$

Power Rule ($n = 1$)

$$= \frac{3}{2}x^2 + C$$

Simplify.

So, the antiderivatives of $3x$ are of the form $\frac{3}{2}x^2 + C$, where C is any constant.

the general pattern of integration

Original integral $\Rightarrow$ Rewrite $\Rightarrow$ Integrate $\Rightarrow$ Simplify

Basic Integration Rules

EXAMPLE.

$$\int k \, dx = kx + C$$

$$\int 5 \, dx = 5x + C$$

$$\int -8 \, dx = -8x + C$$

$$\int \beta \, dx = \beta x + C$$

$$\int k f(x) \, dx = k \int f(x) \, dx$$

$$\int x^n \, dx = \frac{x^{n+1}}{n+1} + C, \quad n \neq -1$$

$$\begin{aligned} \int -2x \, dx &= -2 \int x \, dx \\ &= -2 \frac{x^2}{2} = -x^2 + C \end{aligned}$$

$$\int \pi x \, dx = \frac{\pi x^2}{2} + C$$

Basic Integration Rules

EXAMPLE

$$\int x^2 \, dx = \frac{x^3}{3} + C$$

$$\int x^5 \, dx = \frac{x^6}{6} + C$$

$$\int x^9 \, dx = \frac{x^{10}}{10} + C$$

$$\int 3x^2 \, dx = \frac{3x^3}{3} = x^3 + C$$

$$\int 10x^4 \, dx = \frac{10x^5}{5} = 2x^5 + C$$

$$\int 8x^7 \, dx = \frac{8x^8}{8} = x^8 + C$$

EXAMPLE 3 Rewriting Before Integrating

Original Integral	Rewrite	Integrate	Simplify
a. $\int \frac{1}{x^3} \, dx$	$\int x^{-3} \, dx$	$\frac{x^{-2}}{-2} + C$	$-\frac{1}{2x^2} + C$
b. $\int \sqrt{x} \, dx$	$\int x^{1/2} \, dx$	$\frac{x^{3/2}}{3/2} + C$	$\frac{2}{3}x^{3/2} + C$
c. $\int 2 \sin x \, dx$	$2 \int \sin x \, dx$	$2(-\cos x) + C$	$-2 \cos x + C$

Remember that you can check your answer to an antidifferentiation problem by differentiating.

Basic Integration Rules

$$\int \sqrt[4]{x} \, dx = \int x^{1/4} \, dx = \frac{x^{5/4}}{5/4} = \frac{4}{5}x^{5/4}$$

$$\int \frac{1}{x} \, dx = \ln|x| + C$$

$$\int \frac{2}{x} \, dx = 2 \ln|x| + C$$

$$\int \frac{5}{x} \, dx = 5 \ln|x| + C$$

Basic Integration Rules

$$\int \frac{1}{x^5} dx \quad ? = \int x^{-5} dx = \frac{x^{-4}}{-4} = -\frac{1}{4x^4} + C$$

$$\int \frac{1}{\sqrt{x}} dx = \int x^{-\frac{1}{2}} dx = \frac{x^{\frac{1}{2}}}{\frac{1}{2}} = 2\sqrt{x} + C$$

$$\int \frac{1}{\sqrt[3]{x}} dx = \int x^{-\frac{1}{3}} dx = \frac{x^{\frac{2}{3}}}{\frac{2}{3}} = \frac{3\sqrt[3]{x^2}}{2} + C$$

BREAK

EXAMPLE 4 Integrating Polynomial Functions

$$\text{a. } \int dx = \int 1 dx = x + C$$

Integrand is understood to be 1.

Integrate.

$$\begin{aligned} \text{b. } \int (x + 2) dx &= \int x dx + \int 2 dx \\ &= \frac{x^2}{2} + C_1 + 2x + C_2 \\ &= \frac{x^2}{2} + 2x + C \end{aligned}$$

Integrate.

$C = C_1 + C_2$

The second line in the solution is usually omitted.

$$\begin{aligned} \text{c. } \int (3x^4 - 5x^2 + x) dx &= \frac{3x^5}{5} - \frac{5x^3}{3} + \frac{x^2}{2} \\ &= \end{aligned}$$

Integrate.

Simplify.

EXAMPLE 5 Rewriting Before Integrating

$$\begin{aligned} \int \frac{x+1}{\sqrt{x}} dx &= ? \int \frac{x}{x^{\frac{1}{2}}} + \frac{1}{x^{\frac{1}{2}}} dx \\ &= \int x^{\frac{1}{2}} + x^{-\frac{1}{2}} dx \\ &= \frac{x^{\frac{3}{2}}}{\frac{3}{2}} + \frac{x^{\frac{1}{2}}}{\frac{1}{2}} \\ &= \frac{2x\sqrt{x}}{3} + 2\sqrt{x} + C \\ &= \end{aligned}$$

Rewrite as two fractions.

Rewrite with fractional exponents.

Integrate.

Simplify.

EXAMPLE 6 Rewriting Before Integrating

$$\int \sec x \tan x \, dx = \sec x + C$$

$$\begin{aligned} \int \frac{\sin x}{\cos^2 x} \, dx &= \int \frac{\sin(x)}{\cos(x)} \cdot \frac{1}{\cos(x)} \, dx \\ &= \int \tan(x) \cdot \sec(x) \, dx \\ &= \sec(x) + C \end{aligned}$$

$$\int 3 \cos x \, dx = 3 \sin(x) + C$$

$$\int \frac{1}{\sin^2 x} \, dx = -\cot(x)$$

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Initial Conditions and Particular Solutions

$$y = \int (3x^2 - 1) \, dx = x^3 - x + C \quad \text{General solution}$$

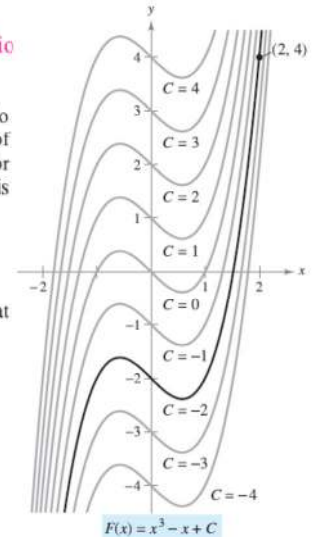
In many applications of integration, you are given enough information to determine a **particular solution**. To do this, you need only know the value of $y = F(x)$ for one value of x . This information is called an **initial condition**. For example, in Figure, only one curve passes through the point $(2, 4)$. To find this curve, you can use the following information.

$$F(x) = x^3 - x + C \quad \text{General solution}$$

$$F(2) = 4 \quad \text{Initial condition}$$

By using the initial condition in the general solution, you can determine that $F(2) = 8 - 2 + C = 4$, which implies that $C = -2$. So, you obtain

$$F(x) = x^3 - x - 2. \quad \text{Particular solution}$$



The particular solution that satisfies the initial condition $F(2) = 4$ is $F(x) = x^3 - x - 2$.

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EXAMPLE 7 Finding a Particular Solution

Find the general solution of

$$F'(x) = \frac{1}{x^2}, \quad x > 0$$

and find the particular solution that satisfies the initial condition $F(1) = 0$.

Solution To find the general solution, integrate to obtain

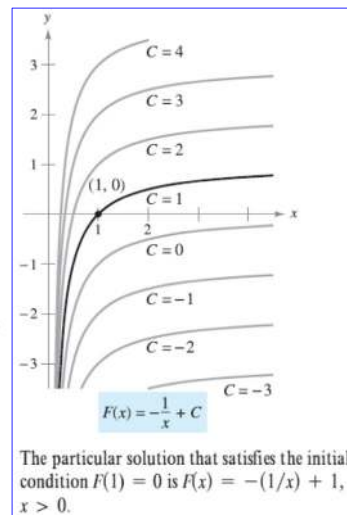
$$\begin{aligned} F(x) &= \int \frac{1}{x^2} \, dx && F(x) = \int F'(x) \, dx \\ &= \int x^{-2} \, dx && \text{Rewrite as a power.} \\ &= \frac{x^{-1}}{-1} + C && \text{Integrate.} \\ &= -\frac{1}{x} + C, \quad x > 0. && \text{General solution} \end{aligned}$$

Using the initial condition $F(1) = 0$, you can solve for C as follows.

$$F(1) = -\frac{1}{1} + C = 0 \Rightarrow C = 1$$

So, the particular solution, as shown in Figure, is

$$F(x) = -\frac{1}{x} + 1, \quad x > 0. \quad \text{Particular solution}$$



The particular solution that satisfies the initial condition $F(1) = 0$ is $F(x) = -(1/x) + 1$, $x > 0$.

EXAMPLE 8 Solving a Vertical Motion Problem

A ball is thrown upward with an **initial velocity of 64 feet per second** from an **initial height of 80 feet**.

- Find the position function giving the height s as a function of the time t .
- When does the ball hit the ground?

Solution

- Let $t = 0$ represent the initial time. The **two given initial conditions** can be written as follows.

$$s(0) = 80 \quad \text{Initial height is 80 feet.}$$

$$s'(0) = 64 \quad \text{Initial velocity is 64 feet per second.}$$

Using -32 feet per second per second as the acceleration due to gravity, you can write

$$s''(t) = -32$$

$$s'(t) = \int s''(t) \, dt = \int -32 \, dt = -32t + C_1.$$

Using the initial velocity, you obtain $s'(0) = 64 = -32(0) + C_1$, which implies that $C_1 = 64$. Next, by integrating $s'(t)$, you obtain

$$s(t) = \int s'(t) \, dt = \int (-32t + 64) \, dt = -16t^2 + 64t + C_2.$$

Using the initial height, you obtain

$$s(0) = 80 = -16(0^2) + 64(0) + C_2$$

EXAMPLE 8 Solving a Vertical Motion Problem

A ball is thrown upward with an initial velocity of 64 feet per second from an initial height of 80 feet.

- Find the position function giving the height s as a function of the time t .
- When does the ball hit the ground?

Solution

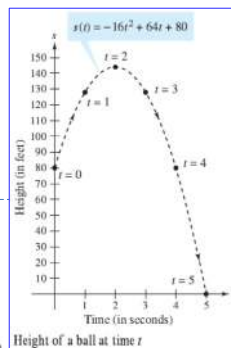
which implies that $C_2 = 80$. So, the position function is

$$s(t) = -16t^2 + 64t + 80.$$

- Using the position function found in part (a), you can find the time at which the ball hits the ground by solving the equation $s(t) = 0$.

$$\begin{aligned} s(t) &= -16t^2 + 64t + 80 = 0 \\ -16(t+1)(t-5) &= 0 \\ t &= -1, 5 \end{aligned}$$

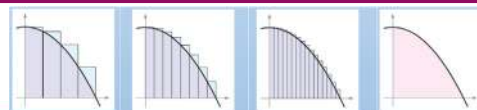
Because t must be positive, you can conclude that the ball hits the ground 5 seconds after it was thrown. ■



Original Integral	Rewrite	Integrate	Simplify
$\int \frac{2}{\sqrt{x}} dx$	$\int \frac{2}{x^{\frac{1}{2}}} dx$	$4\sqrt{x} + C$	
$\int (t^2 + 1)^2 dt$	$\int t^4 + 2t^2 + 1 dt$	$\frac{t^5}{5} + \frac{2t^3}{3} + t + C$	
$\int \frac{x^3 + 3}{x^2} dx$	$\int x + 3x^{-2} dx$	$\frac{x^2}{2} - \frac{3}{x} + C$	
$\int \sqrt[3]{x}(x-4) dx$	$\int x^{\frac{1}{3}}(x-4) dx$	$\frac{x^{\frac{7}{3}}}{\frac{7}{3}} - \frac{4x^{\frac{4}{3}}}{\frac{4}{3}}$	$= \frac{3x^{\frac{7}{3}}\sqrt{x}}{7} - 3x\sqrt[3]{x} + C$

The Area of a Plane Region

II



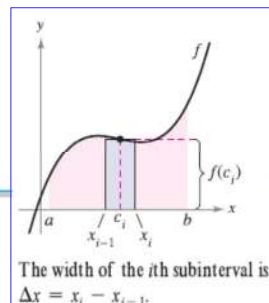
DEFINITION OF THE AREA OF A REGION IN THE PLANE

Let f be continuous and nonnegative on the interval $[a, b]$. The area of the region bounded by the graph of f , the x -axis, and the vertical lines $x = a$ and $x = b$ is

$$\text{Area} = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(c_i) \Delta x, \quad x_{i-1} \leq c_i \leq x_i$$

where $\Delta x = (b - a)/n$

an arbitrary x -value in the i th subinterval



Definite Integrals

Objectives

- Evaluate a definite integral using limits.
- Evaluate a definite integral using properties of definite integrals.

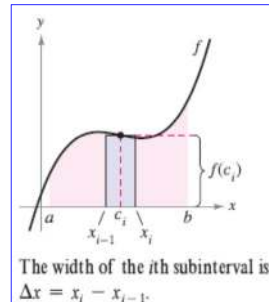
Definite Integrals

$$\lim_{\|\Delta\| \rightarrow 0} \sum_{i=1}^n f(c_i) \Delta x_i = \int_a^b f(x) dx.$$

The limit is called the **definite integral** of f from a to b . The number a is the **lower limit** of integration, and the number b is the **upper limit** of integration.

DEFINITION OF THE AREA OF A REGION IN THE PLANE

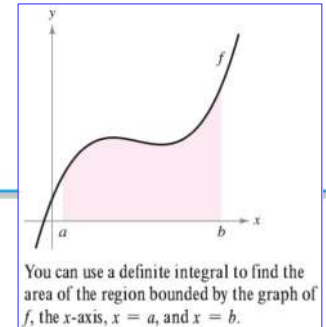
$$\text{Area} = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(c_i) \Delta x, \quad x_{i-1} \leq c_i \leq x_i$$



THEOREM 4.5 THE DEFINITE INTEGRAL AS THE AREA OF A REGION

If f is continuous and nonnegative on the closed interval $[a, b]$, then the area of the region bounded by the graph of f , the x -axis, and the vertical lines $x = a$ and $x = b$ is given by

$$\text{Area} = \int_a^b f(x) dx.$$



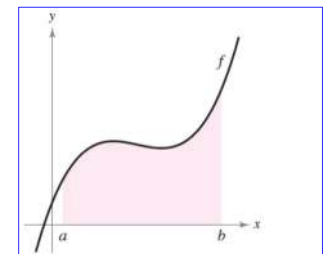
- A **straightforward technique** for evaluating a definite integral will be discussed in the **next lecture**.
- For **now**, however, you can evaluate a definite integral in **two ways**.
 - Use the **limit definition**. OR
 - Use the **area definition** (if the definite integral represents the area of a common geometric region such as a rectangle, triangle, or semicircle).

I will not demonstrate how these two methods are used.

Properties of Definite Integrals

DEFINITIONS OF TWO SPECIAL DEFINITE INTEGRALS

1. If f is defined at $x = a$, then we define $\int_a^a f(x) dx = 0$.
2. If f is integrable on $[a, b]$, then we define $\int_b^a f(x) dx = -\int_a^b f(x) dx$.



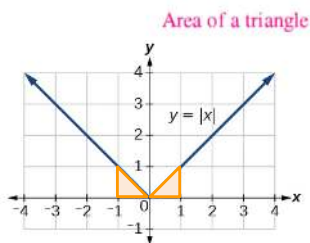
THEOREM 4.6 ADDITIVE INTERVAL PROPERTY

If f is integrable on the three closed intervals determined by a , b , and c , then

$$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx.$$

EXAMPLE 5 Using the Additive Interval Property

$$\begin{aligned}\int_{-1}^1 |x| dx &= \int_{-1}^0 -x dx + \int_0^1 x dx && \text{Theorem 4.6} \\ &= \frac{1}{2} + \frac{1}{2} \\ &= 1\end{aligned}$$



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THEOREM 4.7 PROPERTIES OF DEFINITE INTEGRALS

If f and g are integrable on $[a, b]$ and k is a constant, then the functions kf and $f \pm g$ are integrable on $[a, b]$, and

1. $\int_a^b kf(x) dx = k \int_a^b f(x) dx$
2. $\int_a^b [f(x) \pm g(x)] dx = \int_a^b f(x) dx \pm \int_a^b g(x) dx.$

Note that Property 2 of Theorem 4.7 can be extended to cover any finite number of functions. For example,

$$\int_a^b [f(x) + g(x) + h(x)] dx = \int_a^b f(x) dx + \int_a^b g(x) dx + \int_a^b h(x) dx.$$

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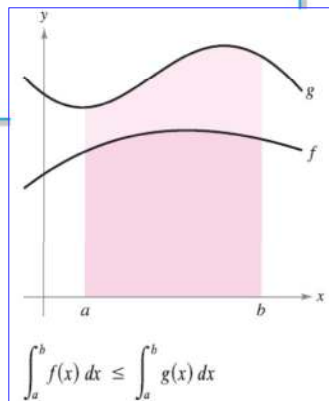
THEOREM 4.8 PRESERVATION OF INEQUALITY

1. If f is integrable and nonnegative on the closed interval $[a, b]$, then

$$0 \leq \int_a^b f(x) dx.$$

2. If f and g are integrable on the closed interval $[a, b]$ and $f(x) \leq g(x)$ for every x in $[a, b]$, then

$$\int_a^b f(x) dx \leq \int_a^b g(x) dx.$$



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Homework

Find the antiderivative of the function below.

1) $f'(x) = 8x^3 + 5\sqrt[3]{x^2} - \frac{10}{x^6} + 7$

2) $g'(x) = \frac{3\cos x}{\sin^2 x}$

3) $y' = \frac{x^2 + 3\sqrt{x}}{x^3}$

- Write down a step-by-step solution in a piece of A4 paper (or on an electronic paper).
- **DO NOT TYPE.**
- Submit your solution in MS Team.
- Do not forget to write your name and student ID.

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